Incremental Detection of Hyperspectral Targets in Consistent Scenes With Continuous Learning

Wenqian Dong, *Member, IEEE*, Yonghui Chen, Song Xiao, Jiahui Qu, *Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— Hyperspectral target detection is a binary classification problem of detecting targets by utilizing the spectral characteristics of hyperspectral images (HSIs). However, in real-world applications, there is more than one class of interest in the same scene. The accuracy of detection for the previously learned target classes may be decreased when the model is retrained for detecting new target classes in scenes, which is called catastrophic forgetting. Consequently, how to ensure that the model has high detection performance for previously learned targets while learning new ones has become a key challenge for hyperspectral multitarget detection tasks. In this article, we propose an incremental detection of hyperspectral targets (IDHTs) method based on continual learning. IDHT decomposes the multitarget detection task into a series of independent subtasks and learns them sequentially. Within each subtask, our proposed incremental spectral detector (ISD) enables the training and learning of new class targets. Simultaneously, we introduce the previous label replay strategy (PLRS), which synthesizes fused labels for the current task training by combining detection outcomes from the previous model with pseudo-labels of the current target. PLRS effectively bridges the knowledge gap across various subtasks in hyperspectral multitarget detection tasks. The proposed IDHT can flexibly and dynamically adapt to new categories and overcome the limitations of fixed-category feature learning. In addition, two hyperspectral datasets are disclosed to evaluate the proposed method. Our method demonstrates significant effectiveness and superiority on both public datasets and two self-collected datasets. Our code and dataset are available at https://github.com/Jiahuiqu/IDHT.**

***Index Terms*— Continual learning, convolutional neural network, hyperspectral image (HSI), muti-target detection.**

Received 11 May 2024; revised 9 August 2024; accepted 11 September 2024. Date of publication 16 September 2024; date of current version 30 September 2024. This work was supported in part by the China Postdoctoral Science Special Foundation under Grant 2022T150508 and Grant 2023T160502; in part by the National Natural Science Foundation of China under Grant 62101414, Grant 62201423, Grant 62471359, and Grant 62476013; in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 959202313052; in part by the Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20230117; in part by the China Postdoctoral Science Foundation under Grant 2021M702548 and Grant 2021M702546; in part by the Youth Innovation Team of Shaanxi Universities; and in part by the Interdisciplinary Cultivation Program of Xidian University under Grant 21103240012. *(Corresponding authors: Song Xiao; Jiahui Qu.)*

Wenqian Dong, Yonghui Chen, Jiahui Qu, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: wqdong@xidian.edu.cn; chenyonghui@stu.xidian.edu.cn; jhqu@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Song Xiao is with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China, and also with Beijing Electronic Science and Technology Institute, Beijing 100070, China (e-mail: xiaosong_xd@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3461868

NOMENCLATURE

Notion	Description
$\mathbf{X} \in \mathbf{R}^{w \times h \times c}$	Hyperspectral image.
$\mathbf{B}_t$	Background pixels for task t.
$\mathbf{Y}_t$	Target spectrum for task t.
$y^{t-1}(h, w, t)$	Results label for task $t-1$.
$\widehat{y}^t(h, w, 2)$	Pseudo label for task t.
$\widetilde{y}^t(h, w, t+1)$	Fusion label for task t.
h	Spatial height.
w	Spatial width.
c	Number of spectral bands.
t	Number of new targets.

I. INTRODUCTION

HYPERSPECTRAL image (HSI), owing to its wealth of spectral information, is widely applied in many fields, such as environmental monitoring, mineral exploration, and land cover mapping [1], [2], [3]. Due to the rich spectral features [4], [5], hyperspectral target detection has gained significant research attention. The primary task in hyperspectral target detection is to discriminate targets and background regions in HSI using the spectral attributes of the objects, as shown in Fig. 1 [6], [7], [8]. In essence, the target detection task can be seen as a binary classification problem, wherein image pixels are labeled as either target or background [9], [10], [11].

Early hyperspectral target detection algorithms mainly designed target detectors through spectral matching technology, such as spectral angle mapper (SAM) [12], matched filter (MF) [13], generalized likelihood ratio test (GLRT), and adaptive cosine/coherence estimation (ACE) [14], which are typical target detectors constructed through binary hypothesis testing. Some researchers characterize HSIs with a linear mixing model (LMM), which holds that each pixel can be approximated as a linear combination of several endmembers with varying fractional abundances and that the perceived target endmembers can be enhanced by some constraint. For example, orthogonal subspace projection (OSP) projects the spectral vector into an orthogonal subspace of the background subspace to suppress the background and highlight the target area [15]. Constrained energy minimization (CEM) identifies areas of interest by reducing the influence of the background while amplifying the target signal [16]. The CEM algorithm has attracted the attention of many researchers due to its

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

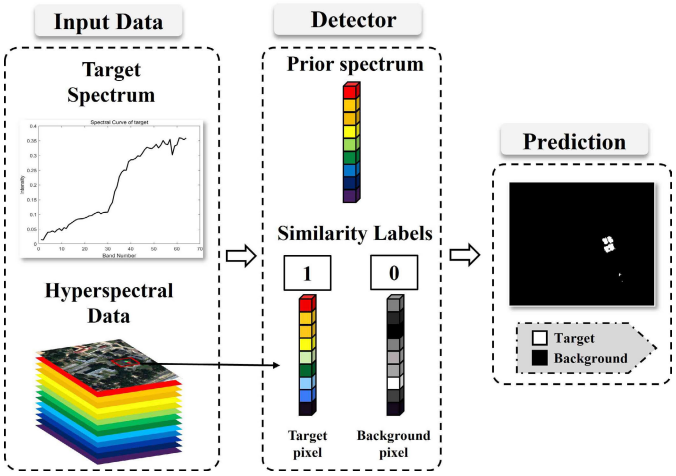


Fig. 1. Schematic of hyperspectral target detection. Prior spectra are used to detect all targets that conform to the spectral curve.

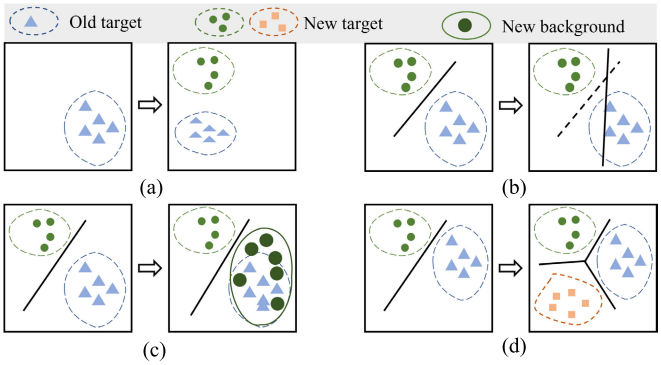


Fig. 2. Three causes of catastrophic forgetting in hyperspectral multitarget detection. (a) Feature shift, (b) classifier distortion, (c) background displacement, and (d) ideal case.

significant advantages in target detection. Zou and Shi [17] introduced the concept of hierarchical CEM (hCEM), which incorporates a series of multilayered CEM to further improve detection accuracy. Recently, machine learning has attracted great attention, and a variety of target detection methods based on machine learning have emerged. For example, Chen et al. [18] propose the sparse-based target detector (STD). Li et al. [19] propose a combined sparse and collaborative representation detector (CSCR). Recently, some deep learning-based target detectors have been proposed for hyperspectral target detection due to the unique learning capabilities of deep learning models. Due to insufficient labeled training samples and an imbalance between target and background classes, the prevailing approach utilizes data augmentation of training samples by synthesizing the target pixels, which in turn serves as the training set to train the end-to-end detector, including convolutional neural network target detector (CNND) [20], a deep convolutional neural network for target detection in hyperspectral imagery (HTD-Net) [21], and two-stream convolutional network-based target detector (TSCNTD) [22]. Lately, some scholars have utilized self-supervised networks to extract spectral features and detect them with matching strategies. For instance, Wang et al. [23] presented self-consistent learning for hyperspectral target detection (SCLHTD), employing an adversarial convolutional autoencoder to capture the characteristics of both odd and even band numbers in the input image.

Despite the significant progress in hyperspectral target detection techniques, most existing methods are designed for specific targets and rely on specific prior spectral information for detection. However, in practical applications, there is often a desire to design methods capable of continuously detecting multiple categories of targets in a single HSI, rather than being limited to a single category. Yet, existing research on hyperspectral multitarget detection tasks remains largely unexplored. Directly applying a network designed for one category of targets to other categories of targets can lead to a decrease in detection performance for the original category of targets, a phenomenon known as catastrophic forgetting. We analyzed the catastrophic forgetting problem caused by incremental detection and identified three main causes, as shown in Fig. 2: 1) feature shift (i.e., the shift in feature extraction caused by changes in the feature extractor); 2) classifier distortion (i.e., the distortion of the weights of the classifier for old classes due to the incremental update of new class samples); and 3) background displacement (i.e., the background samples in the training phase of each new class of targets contain samples of the former class of targets that have been learned, making it difficult for the model to distinguish between the background and the target). The combined effect of these three factors leads to the serious catastrophic forgetting problem in HSI incremental target detection.

Inspired by these, in this article, an incremental detection of the hyperspectral target (IDHT) method for hyperspectral multitarget detection is proposed. Grounded in the concept of continual learning, IDHT transforms the task of multiclass target detection into an incremental learning task, ensuring excellent detection performance for both new and old classes. The model decomposes various target categories into multiple subtasks, which are then trained sequentially for single-target detection. In the incremental learning process for new categories, preserving the memory of previously learned categories poses a challenge due to the absence of training samples for the old categories. Drawing inspiration from the pseudo-label mutual guidance training strategy as described in [24], the model leverages the detection outcomes from the previous model iterations to maintain the memory of old categories. This approach ensures that the detection performance for previously learned categories remains stable and does not degrade over time. In each subtask, our proposed incremental spectral detector (ISD) maps the feature representation of learned targets into an embedding space. By incorporating distillation loss, it aligns the projection space to prevent the offset in the features of previous targets caused by the learning of new classes. Meanwhile, ISD is capable of preserving prior target categories based on the task order, thereby avoiding issues of classifier distortion and bolstering the accuracy and stability of detection. To mitigate the impact of background shift on hyperspectral multitarget detection tasks, we introduced the previous label replay strategy (PLRS). PLRS utilizes detection outcomes from previous training tasks

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

to identify and extract pixels that contain previous label information within the current background class. This extracted information acts as a bridge for knowledge transfer and integration across various hyperspectral multitarget detection subtasks.

Our contributions can be summarized as follows.

1) We propose an IDHT method for multitarget detection tasks in HSIs. IDHT decomposes the task of detecting different categories of targets into multiple subtasks and performs sequential detection of each target in order.
2) We propose the PLRS to utilize the detection results of the old model as pseudo-labels for old category samples to train the new category model, addressing the challenge of background shift.
3) We designed an ISD that overcomes catastrophic forgetting by using the results of PLRS to train the detector to recognize current and previous targets.
4) We have publicly released two real datasets, Chang'an-3 and Chang'an-4, designed for hyperspectral target incremental detection tasks. The dataset is called "Chang'an," which was built by our team in Xi'an, Shaanxi Province. We selected two HSIs from the dataset and annotated multiple categories to verify the effectiveness and practicality of our method.

The rest of this article is structured as follows. Section II presents a review of the relevant methodologies. The proposed method and model are detailed in Section III. Section IV comprises an analysis of the experimental results. Finally, Section V provides a conclusion.

II. Related Work

Continual Learning is a machine learning paradigm that aims to enable models to maintain memory of previous tasks while learning knowledge of new tasks. However, due to the relative independence of data between tasks, the absence of data from the previous task while learning a new task will lead to the performance of the model on the previous task to degrade, a phenomenon known as catastrophic Forgetting [25], [26], [27], [28].

To mitigate catastrophic forgetting, many scholars have proposed some methods, and the existing methods can be classified into four categories: replay-based methods, parameter isolation-based methods, architecture-based methods, and regularization-based methods. Replay-based methods train the model by storing or generating some data from previous tasks and mixing it with data from new tasks to keep the review of old tasks. Some works [29], [30], directly use these data for subsequent training. Rebuffi et al. [31], Douillard et al. [32], and Hu et al. [33] introduce distillation loss to preserve the old knowledge of the model. Parameter-based methods regularize network parameters explicitly [34], [35] or implicitly [36], [37]. Explicit methods aim to prevent drastic changes to important parameters in the old model when learning a new class, while implicit methods are based on knowledge distillation [38]. Architecture-based methods achieve adaptation to different tasks by adding new branches for new tasks or rearranging subnets for specific tasks [39], [40], [41], [42]. Regularization-based methods constrain the change of network weights by adding additional penalty terms to the loss function to prevent forgetting old tasks [38], [43]. In the HSI multitarget detection task, we are not limited by the memory constraints since the training and testing phases are the same remote sensing image. In addition, considering that the replay-based method is the best and simplest to implement for memorizing former classes in continuous learning, we chose the replay-based strategy to solve the catastrophic forgetting problem in the hyperspectral multitarget detection task.

III. Proposed Method

A. Problem Definition and Overview

To address the challenge posed by the diverse categories of targets in real-world scenarios, hyperspectral multitarget detection models need to gradually learn, update, and accumulate information about different targets throughout the detection process. However, this is often constrained by catastrophic forgetting, where learning new targets leads to a sharp decline in the detection performance of old classes. Unlike traditional machine learning models built on the premise of capturing static data distributions [44], [45], [46], continual learning aims to learn from dynamic data distributions and maintain a balanced state of knowledge about new classes [47]. Therefore, we model the task of hyperspectral multitarget detection as a continual learning process aiming at gradually expanding the range of foreground categories and continuously detecting new target categories on unlabeled datasets, while maintaining high-accuracy detection performance on old categories of targets. Through the model, we expect to effectively alleviate the catastrophic forgetting problem and achieve more stable and reliable performance in the task of hyperspectral multitarget detection.

Based on this, we propose an IDHT, which divides the hyperspectral multitarget detection task into initial single-target detection and subsequent incremental detection of new target stages for learning. Each stage comprises a feature extractor and an incremental detector, as shown in Fig. 3. In each learning task for a new category, the model uses a data replay strategy to build a memory space to store the feature information of old categories, and at the same time designs an ISD to learn the feature representation of new categories, thereby overcoming the catastrophic forgetting. Specifically, IDHT divides T categories of targets into T stages and inputs them into the model for training one by one. Since the inputs to each stage are independent of each other, the former class samples are not available during the training of the new class, and the lack of knowledge of the old will make it difficult for the model to maintain a stable detection performance for the old categories. To this end, we propose the PLRS as a bridge for knowledge transfer and integration between the initial learning stage and the incremental learning stage of the hyperspectral multitarget detection task. PLRS generates a fused label that contains both previous and current targets using the detection results of the old model and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

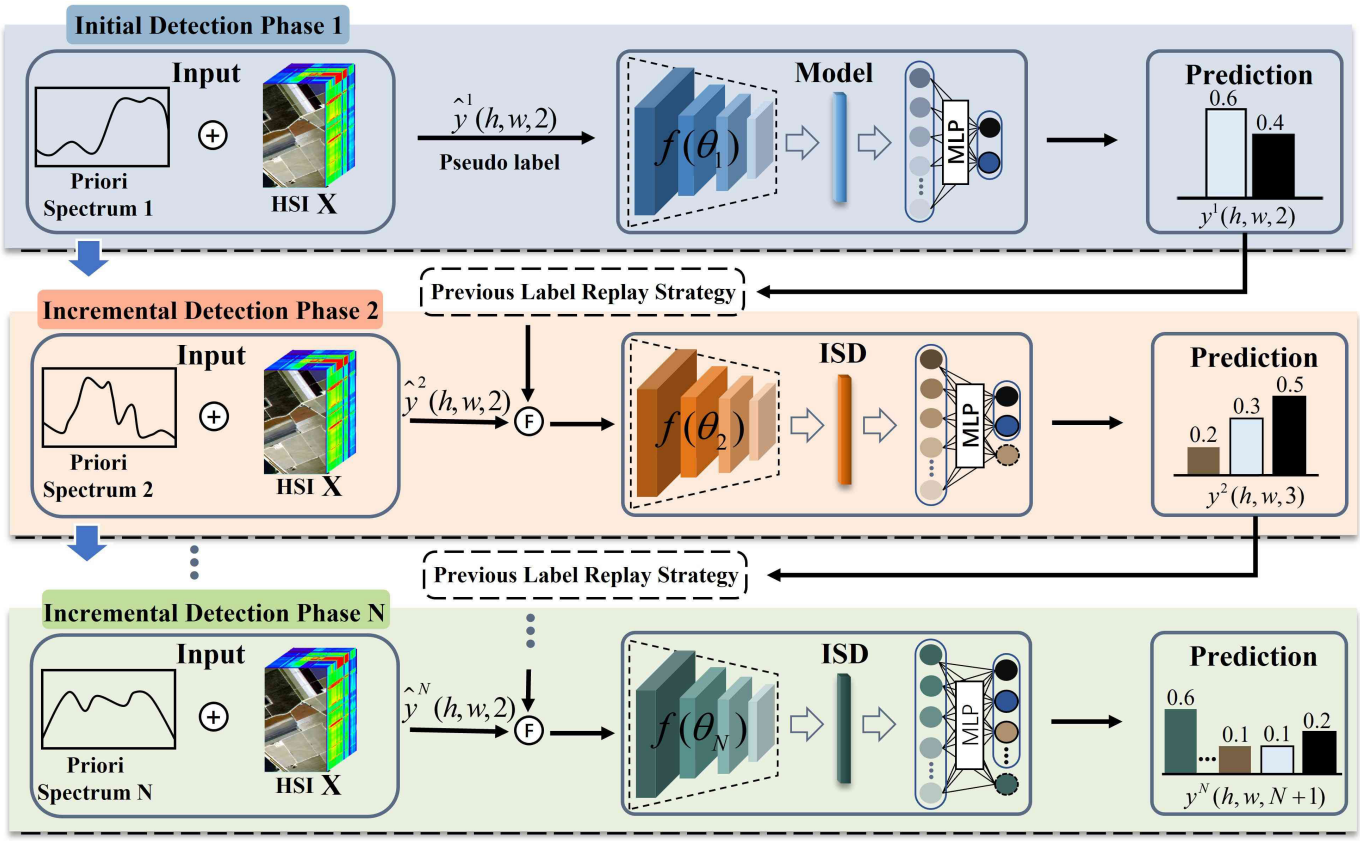


Fig. 3. Overview of the IDHT, which divides the detection phase into an initial detection phase and multiple incremental detection phases. Each phase is equipped with a spectral detector, and the migration of former knowledge between phases is facilitated by PLRS.

pseudo-labels of the current stage, which is used to train the current model, effectively reducing the impact of background shift. In the incremental learning stage, we further propose an ISD to train the model to learn the current task target and old category targets with the ground truth of the PLRS fusion label, avoiding forgetting of old categories. Meanwhile, to maintain the consistency of the feature space between the new and old models, we also introduce distillation loss to avoid the problem of feature drift.

B. Previous Label Replay Strategy

For convenience, we list the key notations and their explanations in this section in Nomenclature. In the initial detection phase of hyperspectral targets, the input of the model consists of HSIs and target spectral curves. The core task is to identify objects in an unlabeled dataset that match the target spectrum. Facing the potential issue of catastrophic forgetting during the incremental detection process of hyperspectral targets, it is particularly important to retain the knowledge of former class targets. Considering that the background class carries a wealth of historical information and potential future information, ignoring these pixels may lead to the loss of valuable information. Since the knowledge of the future is uncertain, what we can rely on is mainly the parameter information provided by the old model. Therefore, we use the output information of the old model as pseudo-labels for the current task, as shown in Fig. 4.

In particular, we use the previous task $t-1$ to predict the samples and obtain the detection results for the old class of targets $y^{t-1} \in \mathbf{R}^{w\times h\times t}$, which is the one-hot encoding, 1 denotes the background, 2 denotes the target $1,\ldots,$ and t denotes the target $t-1$. y^{t-1} contains $t-1$ targets and 1 background, which also represents the shape of the model output. Since the training and test samples of the multitarget detection task for HSIs are the same remote sensing image, we only need to save the detection result $y^{t-1}(h,w,t)$ in the inference of task $t-1$ and do not need to repeat the test at task t. Second, we convert the test results to a ground truth map based on the previous task, where the pixels with a value of $k \in 2,3,\ldots,t$ represent the old-class target. Eventually, we will extract and retain the spectral information $y^{t-1}(h,w,t)$ of the previous target class based on the ground-truth map and the input image $\mathbf{X}$. This facilitates classification training for the current task and ensures that memory for past categories of targets is retained while adapting to training for new categories of targets. The target pseudo-label extracted from the task t can be represented as

$$\widetilde{y}^{t}(h,w,t+1)=\begin{cases}k-1, & \text{if } \mathbf{B}_t=1 \\ & \text{and } x=\operatorname{argmax} y^{k-1}(h,w,k)\\ t, & \text{if } \mathbf{B}_t=0 \text{ and } x=\hat{y}^{t}(h,w,2)\\ 0, & \text{otherwise}\end{cases} \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

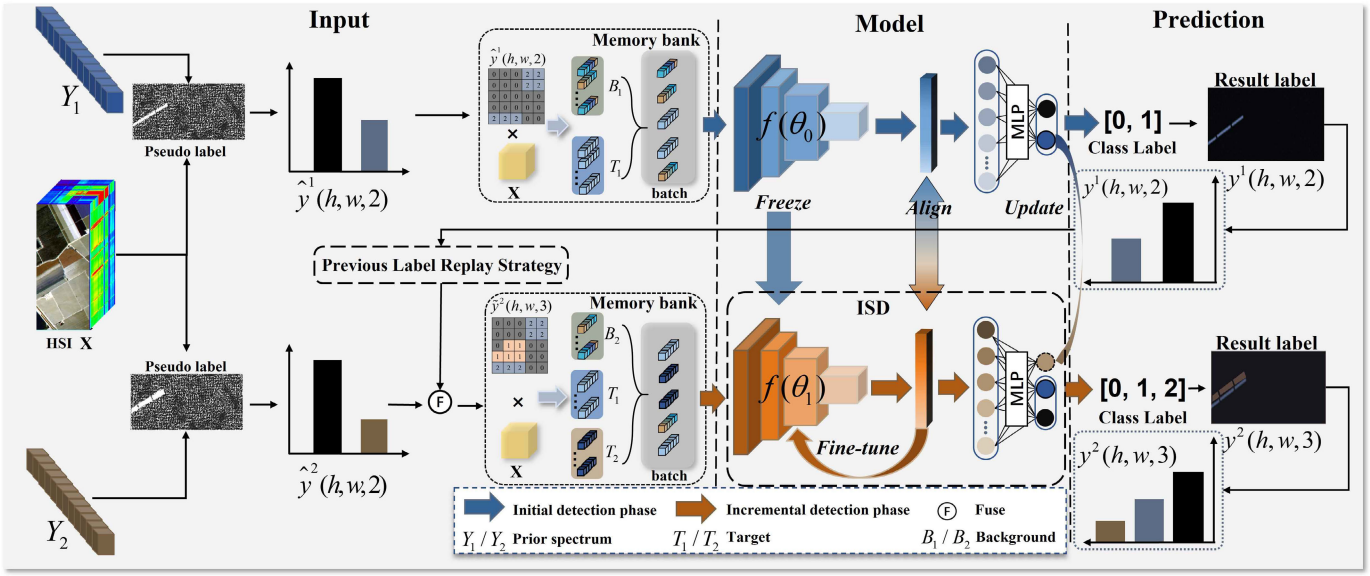


Fig. 4. Detail of IDHT. During the training process, PLRS is constructed based on the data replay, bridging the initial detection stage and the incremental learning stage of the hyperspectral multitarget detection task. Through the output of the old class model, PLRS effectively reduces the influence of background drift. In the incremental learning stage, the proposed ISD utilizes fused labels to retain the memory of old class targets and achieve multitarget detection. At the same time, a distillation loss is introduced to maintain the consistency of the feature space between the new and old models.

where $\mathbf{B}_t = 1$ means that the pixel belongs to the background, $x = \text{argmax} y^{t-1}(h, w, t)$ belongs to the prediction target of step $t-1$, and $x = \hat{y}^t(h, w, 2)$ denotes the pseudo-labeled data sample of the target t. Then, the fusion labels $\tilde{y}^t(h, w, t+1)$ of the old and new targets are stored in the memory bank. Memory bank can obtain the old and new class target samples through these fusion labels, and input these samples into the model according to the batch mixing during the training period, to reduce the blurring between different classes.

The depicted PLRS, as illustrated in Fig. 5, not only augments the availability of pseudo-label information for previously learned class targets in the current task but also reduces ambiguity due to overlap between background and target samples by mixing old and new target samples during training. The strategy of continuously updating the model using different samples also improves the robustness of the model, allowing the model to better handle the different compositions of the background samples and the presence of previously learned targets in them, effectively solving the background bias problem.

C. Incremental Spectral Detector

During the incremental learning phase of hyperspectral targets, the core task lies in how to realize the incremental learning of new classes of targets while maintaining the stable detection performance of old classes. In this regard, we propose an ISD, which effectively balances the model's adaptation to the current new class and the memory stability of the old class, and reduces the distortion of the classifier in the incremental learning of the new class samples through the introduction of distillation loss, discriminat loss, and incremental update strategy.

1) Distillation Loss: Learning discriminative features is crucial for HSI classification [48]. After obtaining the output

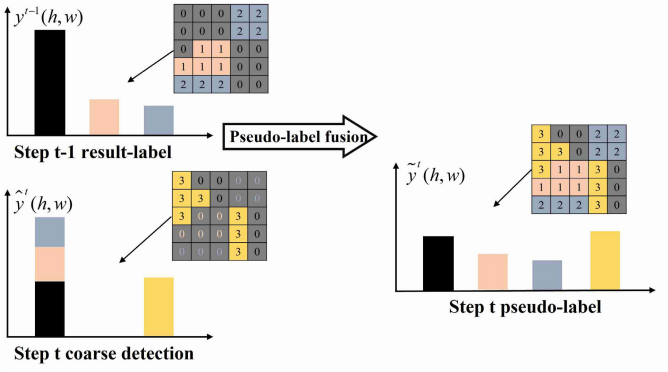


Fig. 5. Illustration of PLRS. We combine the output of the old model on the current data (top left) with the input GT generated by coarse detection of the current step (bottom-left) to obtain the pseudo-labels (middle-right) of the new input GT on the current stage t.

from the previous model by PLRS, it is imperative to sample and train both the target and background, subsequently projecting the target into the latent space. However, the latent spaces of different tasks may vary, leading to confusion in representation and feature drift. To address this, we introduce a distillation loss to align the latent spaces, with the specific formula as follows:

$$\mathcal{L}_d = \text{KL}(P, Q) \tag{2}$$

$$\text{KL}(P, Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)} \tag{3}$$

where $\text{KL}(P, Q)$ is the loss function for KL scatter, P and Q are two probability distributions. In neural networks, (3) is usually expressed using the logarithmic probability form,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

which can be formulated as

$$\mathrm{KL}(P, Q) = \sum_i P(i)(\log P(i) - \log Q(i)). \tag{4}$$

In this article, P and Q denote the softened and logarithmic student model predictions (i.e., $f(\theta_t)$, the feature extractor in the new model) and the soft labeling of the softened teacher model (i.e., $f(\theta_{t-1})$, the feature extractor in the old model), with the specific formula as follows:

$$P = \log(\mathrm{softmax}(f(\theta_t)/\tau)) \tag{5}$$

$$Q = \log(\mathrm{softmax}(f(\theta_{t-1})/\tau)) \tag{6}$$

where τ is the temperature parameter used to adjust the soft labels relative to the new model predictions.

In incremental detection learning, since the $f(\theta_{t-1})$ and $f(\theta_t)$ use the same network structure, this knowledge distillation is closer to self-distillation, which minimizes the difference between the output of the model at stage $t-1$ and the model output at stage t. That is, the latent space of the model mapping is aligned to learn rich, transferable features, which can avoid feature shifts during the learning of a new target.

2) Discriminant Loss: Minimizing the distillation loss maintains the consistency of latent space between the current and previous tasks, mitigating the feature drift of the old classes. However, since the new class of targets is trained with the background as the label in the old model, it leads to confusion in the new class of target feature representation in the current model.

Therefore, we acquire the prior spectral signal from the database first, represented by $\mathbf{Y} = \{\mathbf{Y}_i | i = 1, 2, 3, \ldots\}$, where $\mathbf{Y}_i \in \mathbf{R}^d$, d is the spectral dimension. Given an HSI $\mathbf{X} \in \mathbf{R}^{h\times w\times c}$ and a priori spectral signal $\mathbf{Y}_i$, using the classical method CEM to obtain the pseudo-label of the target $\mathbf{Y}_t$, and then, PLRS is used to generate a training set, which is used to train the detector. An incremental detection classifies the target using a multilayer perceptron (MLP)-based model, which consists of two fully connected layers and a normalization layer. We train the decision boundary of the classifier in the latent space by means of fusion labels $\widetilde{y}^t(h, w)$ generated by PLRS and use cross entropy as a formula for discriminatory loss

$$\mathcal{L}_{\mathrm{ce}}((f(\theta_t, x), y); \mathbf{Y}) = -\log p(x \in \mathbf{Y}_t | f(\theta_t, x)) \tag{7}$$

where x represents training spectrum, $f(\theta_t, x)$ represents a spectral feature in latent space, and $\mathbf{Y}_t$ represents target spectrum for task t

$$x \in \text{class} \quad \arg\max_{i=1}^{T} f(\theta_i, x). \tag{8}$$

3) Incremental Update Strategy: In multitarget detection learning, the decision boundary often shifts unreasonably due to the continuous emergence of new class targets and the forgetting of old ones. This shift not only affects the classifier's ability to recognize new targets but may also lead to confusion with old targets. To mitigate classifier confusion, we propose an architecture-based incremental update strategy that ensures the effective detection of new categories without compromising the recognition capabilities of the former class, thus achieving a balance between the new and old classes

Algorithm 1 Incremental Detection Training Algorithm

input : A priori spectrum $\mathbf{Y}_t$, the HSI $\mathbf{X}$ and a feature extractor parameterized as θ_{t-1}.
output: A result of all targets $y^t(h, w, t+1)$ and a feature extractor parameterized as θ_t.

Importing feature extractor parameters θ_{t-1} into the current model;
Fusion labels $\widetilde{y}^t(h, w, t+1)$ generated from pseudo-labels $\widehat{y}^t(h, w, 2)$ of new class targets with old class result labels $y^{t-1}(h, w, t)$ using PLRS;
for *epoch* $n = 1, 2, \ldots$ **do**
  Update θ_t: minimize distillation loss $\mathcal{L}_d$ to maintain the latent spatial θ_t consistency of old and new potentials;
  Train $\widetilde{y}^t(h, w, t+1)$: minimize $\mathcal{L}_{ce}$ to train ISD classifying targets belonging to former and new classes $\widetilde{y}^t(h, w, t+1)$;
  Renew $y^t(h, w, t+1)$: renew the prediction on result label $y^t(h, w, t+1)$.
end

$$\boldsymbol{f}_i = f(\theta_{t-1}, \mathbf{Y}_i), \quad i = 1, 2, \ldots, t-1 \tag{9}$$

$$\boldsymbol{f}_i = \left[f_i^1, \ldots, f_i^j, \ldots, f_i^L\right] \tag{10}$$

$$\boldsymbol{f}_i = f(\theta_t, \mathbf{Y}_t) \tag{11}$$

$$\boldsymbol{f}_t = \left[f_t^1, \ldots, f_t^j, \ldots, f_t^L\right]. \tag{12}$$

Specifically, we construct a flexible classifier architecture based on MLP. The feature extractor $f(\theta_{t-1})$ can extract ith feature vector $\boldsymbol{f}_i$ corresponding to the original class, as shown in (7) and (8). When new class data $\mathbf{Y}_t$ arrive, the feature extractor $f(\theta_t)$ based on distillation loss freezing can extract the feature vector $\boldsymbol{f}_t$ of the new class, as shown in (9) and (10). The number of neurons in the output layer of the detector is the same as the number of classes in the current task and increases dynamically with the learning of new categories

$$\mathbf{W}_t = \begin{bmatrix} W_1^1 & \cdots & W_i^1 & \cdots & W_{t-1}^1, & W_t^1 \\ \vdots & \ddots & & & \vdots & \vdots \\ W_1^j & & W_i^j & & W_{t-1}^j, & W_t^j \\ \vdots & & & \ddots & \vdots & \vdots \\ W_1^L & \cdots & W_i^L & \cdots & W_{t-1}^L, & W_t^L \end{bmatrix}. \tag{13}$$

The weight column is crucial for precise detection. The updated weights $\mathbf{W}_t$ of the detector not only facilitate the detection of new classes using pseudo-labels but also enable the identification of original class feature vectors using the fused labels generated by PLRS. This design allows the classifier to adjust its structure according to evolving task

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

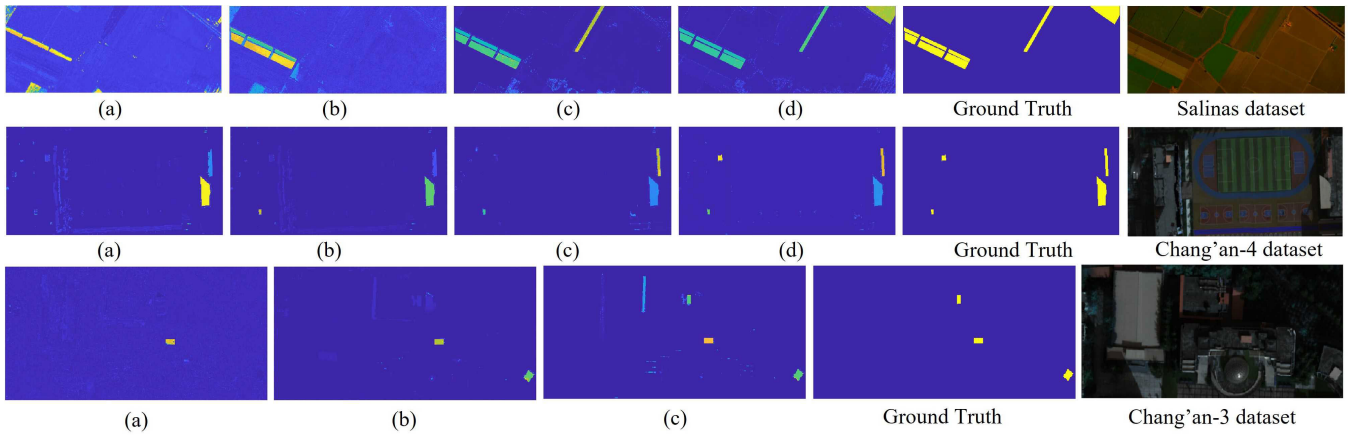


Fig. 6. Detection maps for IDHT applied to different datasets. (a) Test for task 1. (b) Test for task 2. (c) Test for task 3. (d) Test for task 4.

requirements

$$\text{Score}_i = \left[\boldsymbol{f}_i \cdot \mathbf{W}_1, \ldots, \boldsymbol{f}_i \cdot \mathbf{W}_i, \ldots, \boldsymbol{f}_i \cdot \mathbf{W}_{t-1}, \boldsymbol{f}_i \cdot \mathbf{W}_t\right]^T \tag{14}$$

$$\max[\text{Score}_i] = \text{Score}_i[i] = \boldsymbol{f}_i \cdot \mathbf{W}_i \tag{15}$$

$$\text{Score}_t = \left[\boldsymbol{f}_t \cdot \mathbf{W}_1, \ldots, \boldsymbol{f}_t \cdot \mathbf{W}_i, \ldots, \boldsymbol{f}_t \cdot \mathbf{W}_{t-1}, \boldsymbol{f}_t \cdot \mathbf{W}_t\right]^T. \tag{16}$$

During the training process, the Score_i given by the incremental detector is obtained by multiplying the feature vector $\boldsymbol{f}_i$ of ith original class with the updated weights $\mathbf{W}_t$, and $\text{Score}_i[i] = \boldsymbol{f}_i \cdot \mathbf{W}_i$ should be the maximum item in Score_i. When the feature vector $\boldsymbol{f}_t$ of the new class arrives, the incremental detector should give Score_t, as shown in (14), by multiplying the new feature vector with the updated weights $\mathbf{W}_t$, and update the training target with this maximized $\text{Score}_t[t] = \boldsymbol{f}_t \cdot \mathbf{W}_t$. We utilize both discriminative and distillation loss functions, incorporating targets from both new and old categories into the detector. By minimizing the loss function, we reinforce the distinguishability among all categories, thereby bolstering the classifier's overall performance.

4) Inference: During the inference phase of ISD, first, a trained feature extractor network $f(\theta_T)$ is used to take out the spectral signals x to be measured from the input image and map them into the potential space to obtain their feature representations $f(\theta_T, x)$. Then the incremental detector is used to classify the feature representations directly, and the classification results are related to the tasks.

Algorithm 1 summarizes the training process of ISD in the incremental learning phase. The advantage of ISD is that it does not require complex pre-processing or post-processing of the input image, but directly utilizes the priori spectral signal as a reference, simplifying the detection process. The fusion labels generated by PLRS serve as the training set for the new targets, facilitating knowledge transfer between the old and new models, thereby enhancing the stability of the old class and the adaptability of the new class.

D. Training Objective

The IDHT framework defines a feature mapping network and an incremental detector, including distillation loss for spatial alignment and cross-entropy loss for multiclassification

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \lambda * \mathcal{L}_d \tag{17}$$

where λ is a learnable parameter.

IV. Experiment Results and Analysis

A. Dataset

The proposed method is evaluated on three real HSI datasets, including Chang'an-3, Chang'an-4, and Salinas. These datasets are acquired by different sensors, including the airborne visible/infrared imaging spectrometer (AVIRIS) sensor and the hyperspectral digital image collection experiment (HYDICE) onboard sensor in different scenarios. Different categories of targets are distributed in these datasets. Here are some brief introductions to the dataset.

1) Chang'an-3 and Chang'an-4: To evaluate the ability of the proposed method to be applied in real-world scenarios, our team worked together to establish a dataset named Chang'an in Chang'an District, Xi'an, Shaanxi Province. The Chang'an dataset contains Chang'an-3 and Chang'an-4 of size 189 × 398 × 130 and 192 × 398 × 130 collected by Headwall VNIR/SWIR full-band hyperspectrometer, and the four categories of Chang'an-4 and three categories of Chang'an-3 are labeled, respectively, the pseudo-color map is shown in the second and third rows of the final image of Fig. 6.

2) Salinas: This dataset was captured by the AVIRIS sensor in Salinas Valley, California, with a dataset size of 512 × 217 and a spatial resolution of 3.7 m. It contains 224 consecutive bands, and by removing the 20 absorption bands (108–112, 154–167, and 224), the actual number of bands used for training is 204. In this article, four target categories are selected to verify the effect of the model, the pseudo-color map is shown in the first row of the final image of Fig. 6.

B. Experiments Setup

1) Evaluation Metrics: To evaluate the performance of IDHT, we conducted a succinct assessment using the receiver operating characteristic (ROC) curve and its corresponding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

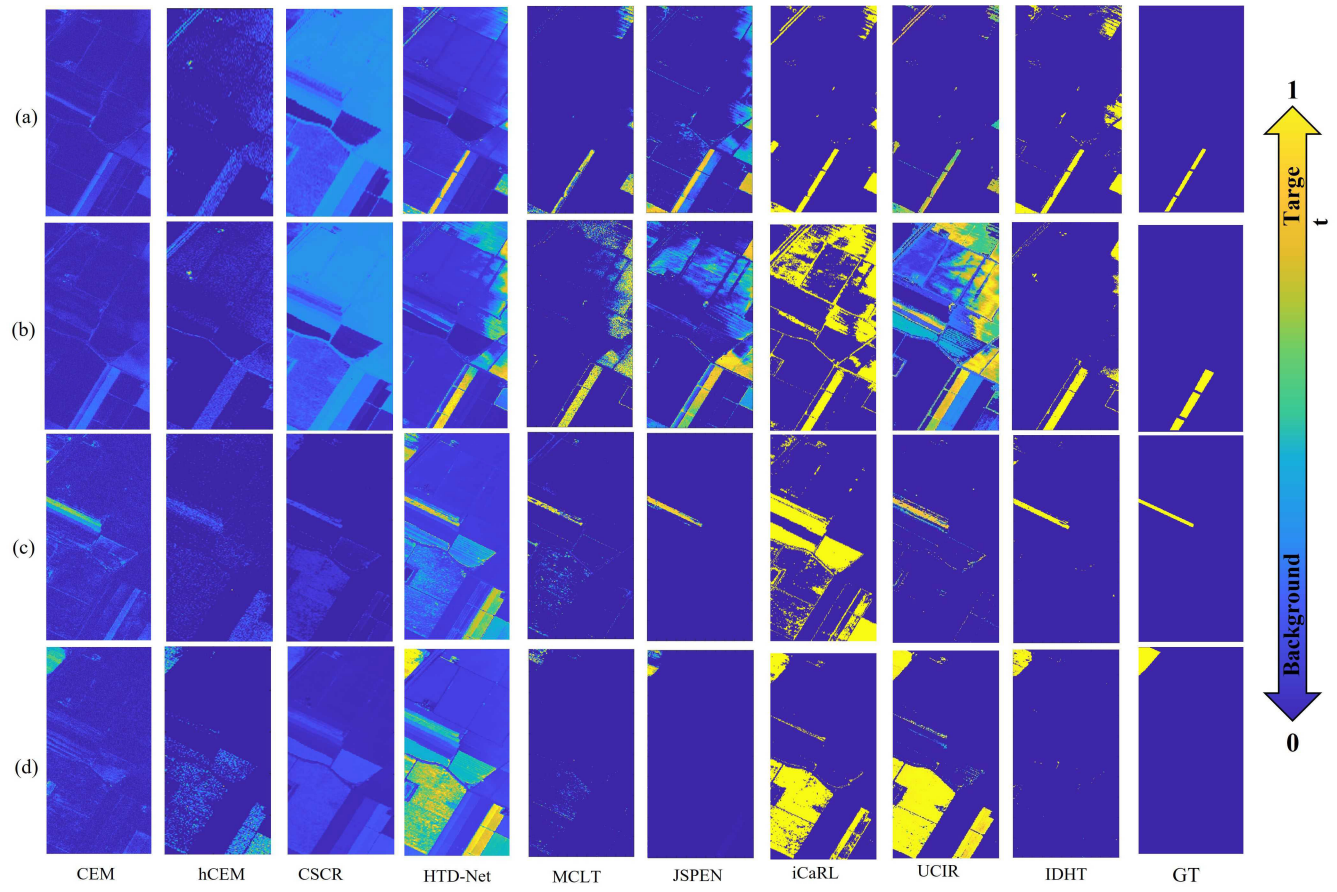


Fig. 7. Detection maps for different methods applied to the Salinas. (a) Test for task 1. (b) Test for task 2. (c) Test for task 3. (d) Test for task 4.

TABLE I

AUC OF ALL THE COMPETING METHODS ON THE THREE DATASETS. THE BEST AND THE SECOND-BEST VALUES ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method \ Dataset		Salinas				Chang'an-4				Chang'an-3		
		target 1	target 2	target 3	target 4	target 1	target 2	target 3	target 4	target 1	target 2	target 3
CEM		0.9324	0.8547	0.9413	0.8941	0.9725	0.9929	0.7546	0.9046	0.9864	0.9637	0.9396
hCEM		0.9521	0.9187	0.8933	0.9057	0.9965	0.9837	0.8983	0.6843	0.9953	0.9918	0.9794
CSCR		0.9749	0.9390	0.8314	0.8112	0.9967	0.6716	0.7525	0.7159	0.9987	0.9548	0.9494
HTD-Net		0.9668	0.9388	0.9670	0.9325	0.9910	0.7636	0.8136	0.9923	0.9712	0.9931	0.9885
MCLT		0.9740	0.9610	0.9124	0.8978	**0.9988**	0.9823	0.8600	0.9983	0.9894	**0.9991**	0.9967
JSPEN		0.9446	0.9657	**0.9794**	**0.9649**	0.9979	0.9833	0.9074	0.9800	0.9473	0.9875	0.9863
iCaRL		0.9722	0.9140	0.9114	0.9311	0.9312	0.7039	0.8874	0.9178	0.9340	0.9718	0.9696
UCIR		0.9552	0.9056	0.9680	0.9203	0.9514	0.8694	0.9333	0.9341	0.9751	0.9912	0.9334
IDHT	task4	0.9797	0.9762	0.9792	0.9647	0.9972	0.9990	0.9542	0.9998	–	–	–
	task3	0.9872	0.9801	0.9718	–	0.9984	0.9994	0.9531	–	0.9981	0.9996	0.9994
	task2	0.9888	0.9815	–	–	0.9986	0.9997	–	–	0.9995	0.9977	–
	task1	0.9858	–	–	–	0.9987	–	–	–	0.9993	–	–
Average of IDHT		**0.9854**	**0.9793**	0.9755	0.9647	0.9982	**0.9994**	**0.9537**	**0.9998**	**0.9990**	0.9987	**0.9994**

area under the curve (AUC). The ROC, a widely accepted metric for target detection, illustrates the relationship between true positive rate (TPR) and false positive rate (FPR) across varying thresholds τ based on the reference map. The AUC, representing the entire area under the ROC curve, serves as a comprehensive measure of detection performance. In this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

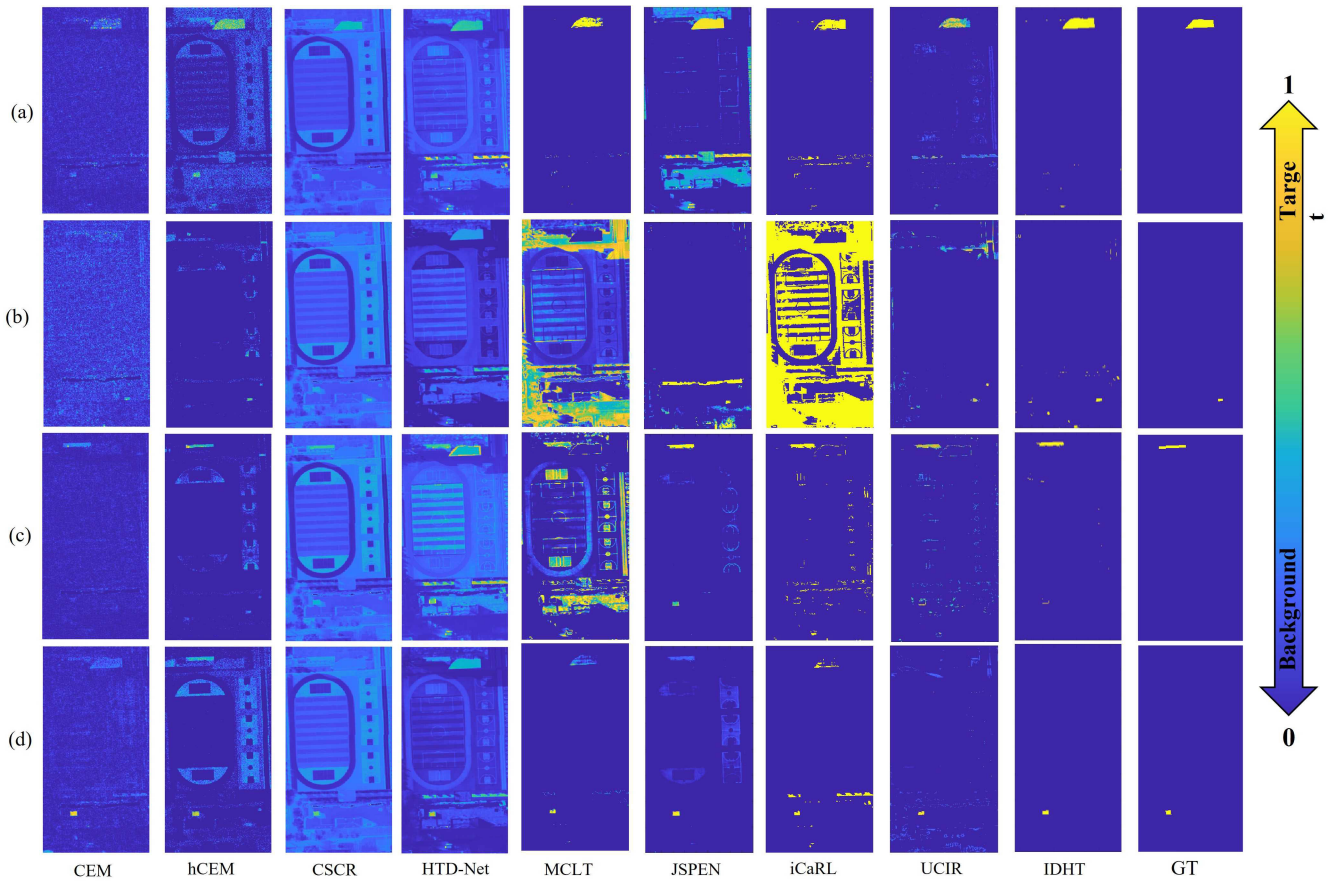


Fig. 8. Detection maps for different methods applied to the Chang'an-4. (a) Test for task 1. (b) Test for task 2. (c) Test for task 3. (d) Test for task 4.

article, AUC scores were used as a way to judge the detection effectiveness of different methods. The better the performance of a detector, the larger the AUC.

2) Implementation Details: The proposed method is implemented on the PyTorch platform, trained using the Adam optimizer on two NVIDIA GeForce 3090 GPUs. Each task is trained with 600 epochs with a batch size of 200 and a learning rate set to 0.001. Additionally, a CosineAnnealingLR strategy is applied to update the learning rate for all models.

3) Competing Methods: To verify the effectiveness and superiority of our proposed method, we compared it with four single-class hyperspectral target detection algorithms and two incremental learning methods for RGB image classification, including: the classical detection method CEM [16], the improved method hCEM for CEM [17], the representation-based target detector CSCR [19], the deep convolutional neural network for target detection in hyperspectral imagery (HTD-Net) [21], the target detector based on unsupervised momentum contrastive learning (MCLT) [49], an interpretable of hyperspectral target detection networks (JSPEN) [8], as well as incremental classifier and representation learning (iCaRL) [31], and learning a unified classifier incrementally via rebalancing (UCIR) [50].

In IDHT, considering that the pseudo-labels of the new class may contain erroneous samples, we randomly select 50% of the sample size of the new class among them for training, and at the same time, we consider the detection results of the former class as high-confidence pseudo-labels, and select 80% of the sample points of the old for co-training to improve the stability of the former class.

For the incremental learning method for RGB image classification, there is no background offset problem given that the input is an HSI with complete target pseudo-labeling. Meanwhile, the prediction mechanism mainly relies on the distance metric to identify the class that is most similar to the target spectrum, and the lack of trainable parameters for the classifier results in the algorithm being able to accurately reflect the detection performance of the model only with the AUC metric calculated without missing the unknown target.

For single-target detection algorithms, we conduct individual detections for each target class without examining the detection performance of the previous classes. For instance, task 1 represents the detection outcome for target 1, and task 2 represents the detection outcome for target 2. Regarding the incremental learning approach, we decompose different classes of targets into multiple subtasks and train them successively. Among them, Task 1 in the experiment indicates the detection results for target 1, which is analogous to the traditional hyperspectral single-target detection task. Task 2 indicates the detection results for target 2 and target 1. For the proposed method, we documented the detection results at each training stage between Task 1 and Task 4 to verify the memorability of the method for the old class of targets and the plasticity of the new class of targets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

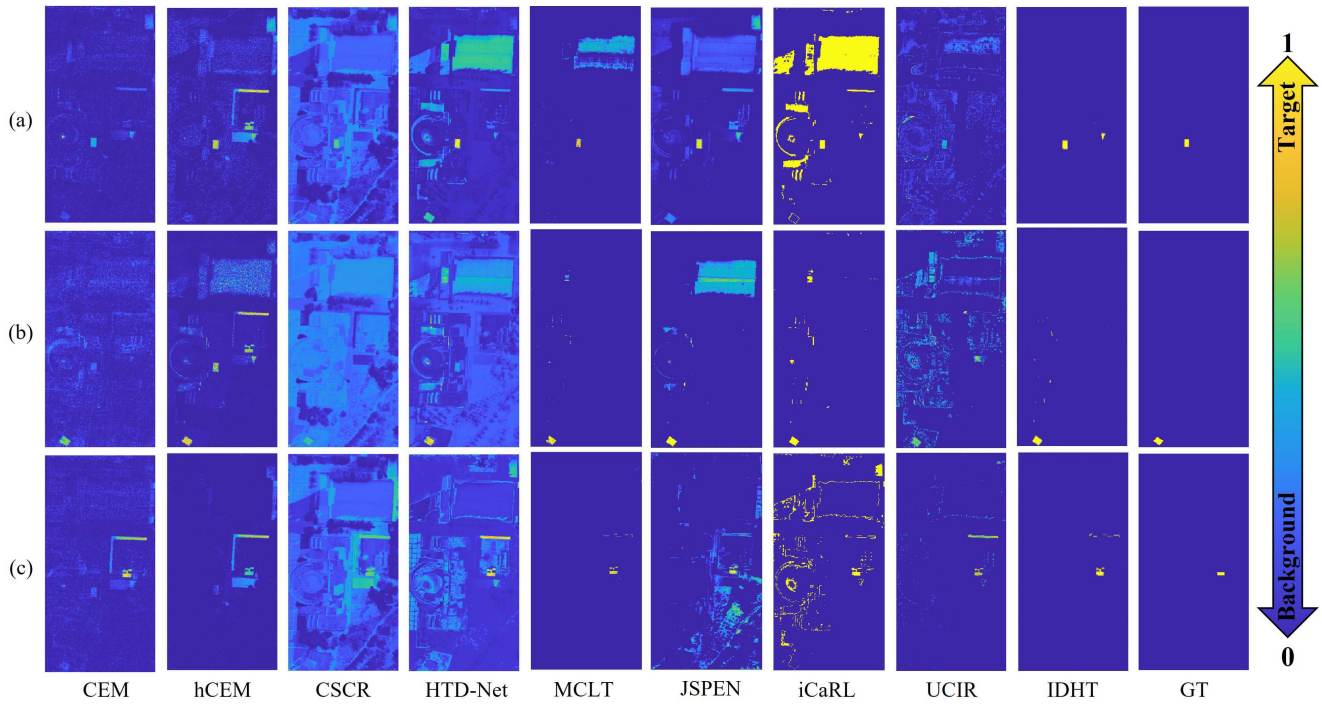


Fig. 9. Detection maps for different methods applied to the Chang'an-3. (a) Test for task 1. (b) Test for task 2. (c) Test for task 3.

C. Performance Comparison

For a fair comparison, the different comparison methods use the same initial training set. The comparison results and subjective plots of the various methods on the benchmark dataset are shown in Table I and Figs. 7–9.

1) Experiment on the Salinas: Table I shows the experimental results. The deep learning method outperforms the classical methods. In the learning of target 4, the detection accuracy of IDHT for target 1 is higher than other methods, and the AUC is only reduced by 0.9039%, while the AUC of target 2 is reduced by 1.2%. The AUC metrics of target 3 and target 4 are basically close to JSPEN, which has the best detection effect. Fig. 7 shows the subjective images. The background suppression effects of CSCR, HTD-Net, iCaRL, and UCIR are much lower than those of other methods. Among them, since iCaRL is based on representation learning for detection, it can be seen that the false detection rate of target 2 is relatively high. Further, due to the more obvious miss-detection of target 4 of MCLT, it results in a relatively poor AUC index. Although both CEM and hCEM can suppress the background output to a very low level, they have different degrees of false positives and false negatives, especially CEM. In contrast, the proposed IDHT not only distinguishes between the target and the background of the task at hand more efficiently but also ensures an excellent detection performance for the old class of target.

2) Experiment on the Chang'an-4: As can be seen from Table I, the results observed by Chang'an-4 are similar to those observed by Salinas, and the detection performance of the classical methods is not as good as that of the deep and incremental learning methods. Among them, the detection performance of the various methods for target 1 basically stays at a high level. However, IDHT learns the new target class better and keeps the forgetting rate of the old class of targets at a low level as the learning task increases, even the metrics of target 3 improve to some extent. As can be seen from Fig. 8, the best background suppression effects in subjective effects are CEM, JSPEN, and IDHT, but there is a certain degree of misdetection of target 3 by CEM and JSPEN, and the background suppression effects of CSCR, HTD-Net, MCLT, and iCaRL on target 2 are worse, and the overall detection performance is not as good as that of IDHT.

3) Experiment on the Chang'an-3: Chang'an-3 has only three target categories. The best detection performance for target 1 and target 3 is IDHT, while the highest detection performance for target 2 is MCLT, but the AUC index of target 2 exceeds that of MCLT in the learning phase of the new target class. As can be seen from the subjective images shown in Fig. 9, the background suppression of CSCR, HTD-Net, JSPEN, and iCaRL is poor, and there is a more In contrast, CEM, hCEM, MCLT, and IDHT have more significant background suppression and target highlighting effects. In contrast, CEM, hCEM, MCLT, and IDHT have more significant effects on background suppression and target highlighting, and IDHT has the best overall performance in terms of AUC and background suppression effect in the subjective map.

D. Ablation Study

In this section, we conduct a series of ablation experiments to evaluate the impact of feature drift, classifier distortion, and background shift in incremental learning. We analyze the impact of the model's feature map network and the incrementally updated classifier on performance. First, to explore the suppression effect of PLRS on background shift, we do

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

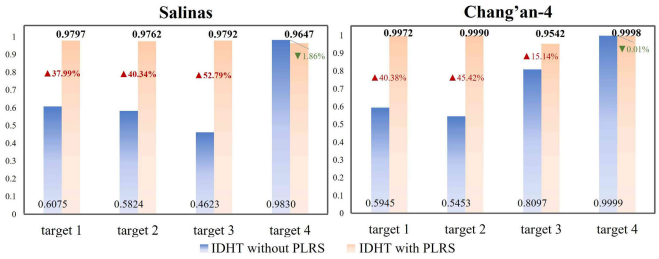


Fig. 10. Detection performance of IDHT with and without PLRS on the two datasets. (Left) Salinas dataset. (Right) Chang'an-4 dataset.

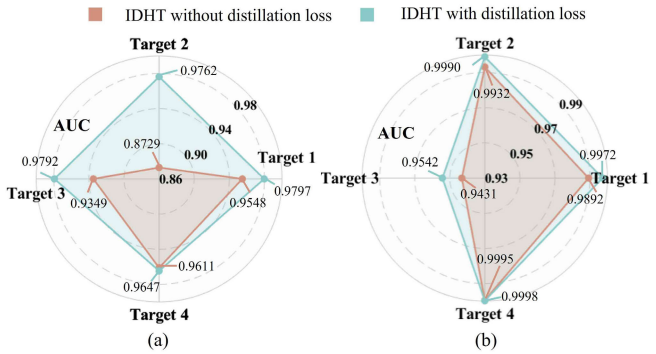


Fig. 11. Detection performance of IDHT with and without distillation loss on the two datasets. (a) Salinas dataset. (b) Chang'an-4 dataset.

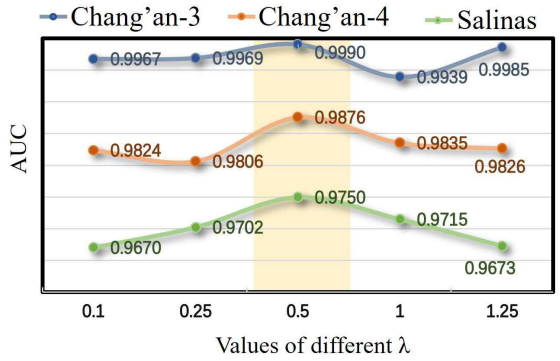


Fig. 12. Detection performance of IDHT with values of different λ on the three datasets.

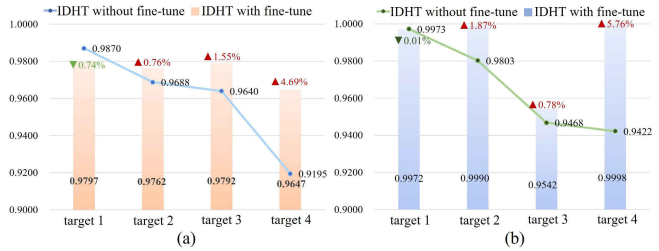


Fig. 13. Detection performance of IDHT with and without fine-tuning on the two datasets. (a) Salinas dataset. (b) Chang'an-4 dataset.

not use PLRS and directly train the new class model to test the detection effect of the detector on the old class spectrum. Second, to evaluate the potential space of the feature map network for overcoming forgetting, we train the respective feature map networks independently without using the distillation loss and test the model's detection effect on the old classes. Additionally, we train the feature map networks with different proportions of distillation loss (i.e., different values of λ) to further assess their potential in mitigating forgetting. Finally, we also verify whether the performance of the model will decrease by freezing the parameters of the feature map network and only training the classifier. To ensure fairness in the experiments, all settings were consistent except for the ablation part, and the detection results from the fourth stage were used as the basis for experimental comparison.

1) Effectiveness of PLRS: To thoroughly investigate the effectiveness of PLRS in suppressing background shift, we trained new class targets directly without employing PLRS and subsequently tested the classifier's detection performance on old class spectra. We conducted the training on the Chang'an-4 and Salinas datasets. As evident from the data presented in Fig. 10, the impact of PLRS on the model is crucial. Although each new class target can achieve satisfactory detection performance, even exhibiting better results on the Salinas dataset, the test metrics for all old class targets mostly hover around 0.5, equivalent to random guessing, indicating the worst possible detector performance. This experimental result validates the effectiveness of PLRS in addressing background shift issues.

2) Influence of Distillation Loss: To further investigate the inhibitory effect of the feature space on feature shift, we trained the respective feature mapping networks independently without using distillation loss, meaning we tested the model's detection performance on old categories without aligning the feature space. As indicated by Fig. 11, in the absence of distillation loss, the model's detection performance for old targets decreased, while its detection performance for new categories became more advantageous. This suggests that aligning the feature space through distillation loss can effectively prevent feature shifts caused by learning new categories, thus enhancing the overall performance and stability of the model. Furthermore, we conducted experiments with varying λ values to investigate the effect of different ratios of distillation loss on the detection performance. As shown in Fig. 12, a λ value of about 0.5 provides the best overall performance on all datasets, with an AUC of 0.9990 for Chang'an-3, 0.9876 for Chang'an-4, and 0.9750 for Salinas. In (17), λ controls the distillation loss, which is designed to balance the detection performance of the old class of targets. It shows that this value of λ effectively balances the effects of the old and new categories of targets, thus enhancing the detection ability and robustness of the model.

3) Influence of Network Parameter Freezing: To validate the classification performance of the classifier for targets, we uniformly employed the encoder from task 1, froze its relevant parameters, and only trained the classifier to test the model's detection effectiveness on old categories. According to the experimental results presented in Fig. 13, we observed that the detection performance of target 1 exhibited good stability. This indicated that the decision boundaries learned by the classifier in the latent space were more inclined toward target 1. However, while the detection performance for newly added targets gradually improved with training time, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

detection performance for new categories deteriorated, leading to a decrease in the model's plasticity.

V. Conclusion

In this article, we propose an IDHT method, to maintain the memory of former classes while learning new classes, which solves the difficulties and challenges of multitarget detection in the same scene. To prevent background shift, our proposed PLRS generates a fused label containing multiple target classes by fusing the output of the old model with the pseudo-labels of the current stage, which is used to train the current model. Then, ISD is designed to train the model to learn both new and old target classes simultaneously, solving the catastrophic forgetting problem. In addition, we also introduce a distillation loss to maintain the consistency of the feature space between the new and old models, avoiding the feature drift problem. The effectiveness and superiority of our method are verified on three real datasets.

References

[1] L. Fang, Y. Jiang, Y. Yan, J. Yue, and Y. Deng, "Hyperspectral image instance segmentation using spectral–spatial feature pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502613.

[2] W. Dong, S. Liu, S. Xiao, J. Qu, and Y. Li, "ISPDiff: Interpretable scale-propelled diffusion model for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519614.

[3] W. Dong, Y. Yang, J. Qu, S. Xiao, and Y. Li, "Local information-enhanced graph-transformer for hyperspectral image change detection with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509814.

[4] C. Li, W. Sun, J. Peng, and K. Ren, "Deep dynamic adaptation network based on joint correlation alignment for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532312.

[5] J. Qu et al., "Progressive multi-iteration registration-fusion co-optimization network for unregistered hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519814.

[6] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Dept. Elect. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep., 2017. [Online]. Available: https://faculty.eng.ufl.edu/machine-learning/2017/03/classification-label-map-for-muufl-gulfport-released/

[7] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Dept. Elect. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.

[8] W. Dong et al., "Deep spatial–spectral joint-sparse prior encoding network for hyperspectral target detection," *IEEE Trans. Cybern.*, pp. 1–13, 2024, doi: 10.1109/TCYB.2024.3403729.

[9] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[10] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[11] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501513.

[12] P. E. Dennison, K. Q. Halligan, and D. A. Roberts, "A comparison of error metrics and constraints for multiple endmember spectral mixture analysis and spectral angle mapper," *Remote Sens. Environ.*, vol. 93, no. 3, pp. 359–367, Nov. 2004.

[13] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, May 1992.

[14] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[15] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[16] W. Farrand, "Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.

[17] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2015.

[18] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[19] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[20] W. Li, G. Wu, and Q. Du, "Transferred deep learning for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5177–5180.

[21] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[22] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[23] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[24] X. Tai, M. Li, M. Xiang, and P. Ren, "A mutual guide framework for training hyperspectral image classifiers with small data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510417.

[25] R. French, "Catastrophic forgetting in connectionist networks," *Trends Cognit. Sci.*, vol. 3, no. 4, pp. 128–135, Apr. 1999.

[26] A. Robins, "Catastrophic forgetting, rehearsal and Pseudorehearsal," *Connect. Sci.*, vol. 7, no. 2, pp. 123–146, 1995.

[27] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," in *Psychology of Learning and Motivation*, vol. 24. Amsterdam, The Netherlands: Elsevier, 1989, pp. 109–165.

[28] Y. Liu, X. Hong, X. Tao, S. Dong, J. Shi, and Y. Gong, "Model behavior preserving for class-incremental learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7529–7540, Oct. 2023, doi: 10.1109/TNNLS.2022.3144183.

[29] E. Belouadah and A. Popescu, "DeeSIL: Deep-shallow incremental learning," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2018, pp. 151–157.

[30] E. Belouadah and A. Popescu, "IL2M: Class incremental learning with dual memory," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 583–592.

[31] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "iCaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2001–2010.

[32] A. Douillard, M. Cord, C. Ollion, T. Robert, and E. Valle, "PODNet: Pooled outputs distillation for small-tasks incremental learning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 86–102.

[33] X. Hu, K. Tang, C. Miao, X.-S. Hua, and H. Zhang, "Distilling causal effect of data in class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3956–3965.

[34] K. James et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[35] R. Aljundi, F. Babiloni, M. Elhoseiny, M. Rohrbach, and T. Tuytelaars, "Memory aware synapses: Learning what (not) to forget," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 139–154.

[36] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2017.

[37] Y. Liu, S. Parisot, G. Slabaugh, X. Jia, A. Leonardis, and T. Tuytelaars, "More classifiers, less forgetting: A generic multi-classifier paradigm for incremental learning," in *Proc. 16th Eur. Conf. Comput. Vis.*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 699–716.

[38] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[39] J. Yoon, E. Yang, J. Lee, and S. Ju Hwang, "Lifelong learning with dynamically expandable networks," 2017, *arXiv:1708.01547*.

[40] A. A. Rusu et al., "Progressive neural networks," 2016, *arXiv:1606.04671*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.

[41] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7765–7773.
[42] J. Serra, D. Suris, M. Miron, and A. Karatzoglou, "Overcoming catastrophic forgetting with hard attention to the task," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4548–4557.
[43] H. Jung, J. Ju, M. Jung, and J. Kim, "Less-forgetting learning in deep neural networks," 2016, *arXiv:1607.00122*.
[44] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.
[45] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.
[46] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511015.
[47] L. Wang, X. Zhang, H. Su, and J. Zhu, "A comprehensive survey of continual learning: Theory, method and application," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5362–5383, Aug. 2024.
[48] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.
[49] Y. Wang, X. Chen, E. Zhao, C. Zhao, M. Song, and C. Yu, "An unsupervised momentum contrastive learning based transformer network for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9053–9068, 2024.
[50] S. Hou, X. Pan, C. C. Loy, Z. Wang, and D. Lin, "Learning a unified classifier incrementally via rebalancing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 831–839.



Wenqian Dong (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She has been a Visiting Scholar with Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 30 papers in refereed journals and conferences, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Remote Sensing*. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.



Yonghui Chen received the B.E. degree in communication engineering from Nanchang University, Nanchang, China, in 2022. He is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China.

His research interests include hyperspectral image target detection and deep learning.



Song Xiao received the M.S. degree in communication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

From 2006 to 2007, she was with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA, as a Post-Doctoral Researcher. She is currently a Professor with Beijing Electronic Science and Technology Institute, Beijing, China, and also a Professor with the School of Telecommunications Engineering, Xidian University. She has authored more than 80 international journal articles and conference papers. Her research interests include image processing, image compression and coding, joint source-channel coding, multimedia transmission systems over wired/wireless network, and compressed sensing.

Dr. Xiao is a Council Member of Shaanxi Society of Image and Graphics and a member of the IEEE Multimedia Communication Technology Committee and the IEEE Signal Processing Society. She is the Secretary-General of the Image Application in Military and Civilian Integration (IAMCI) Professional Committee of China Society of Image and Graphics.



Jiahui Qu (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was an Exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 30 papers in academic journals and conferences, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Remote Sensing*. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University, in 1999, where he is currently a Professor and also the Director of the Image Coding and Processing Center State Key Laboratory of Integrated Service Networks. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:41:05 UTC from IEEE Xplore. Restrictions apply.